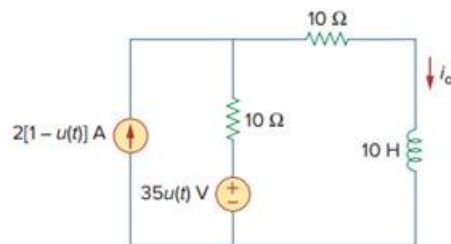


Problem

In the circuit of Fig. 7.143, determine $i_o(t)$.

Fig. 7.143:



Step-by-step solution

Step 1 of 5

(Refer to the Figure 7.143 in the textbook) draw the circuit for time $t < 0$.

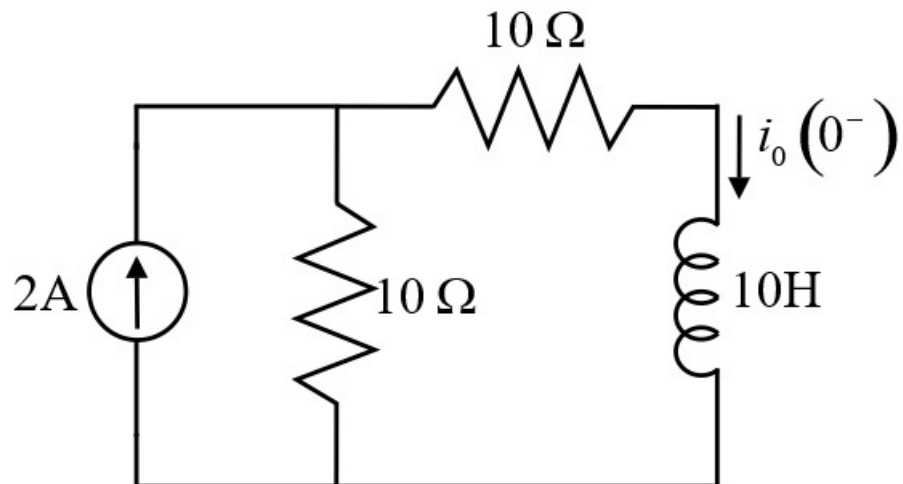


Figure 1

Step 2 of 5

Calculate the current $i(0^-)$ across the inductor.

$$\begin{aligned} i_o(0^-) &= 2 \text{ A} \times \frac{10}{20} \\ &= 1 \text{ A} \end{aligned}$$

Step 3 of 5

Draw the circuit for the time $t > 0$.

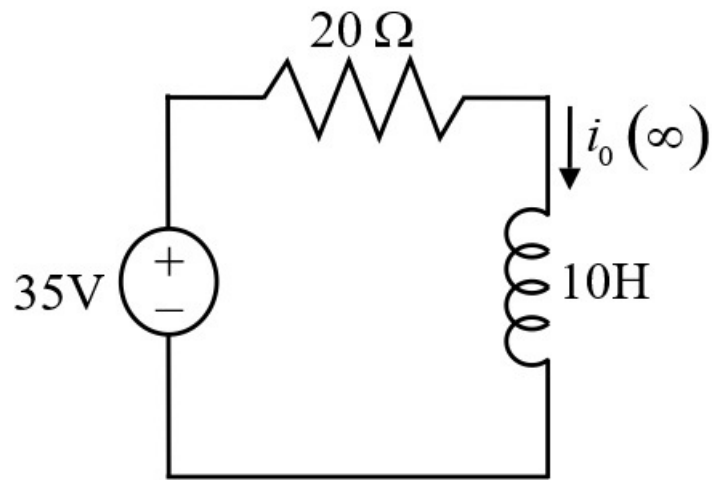


Figure 2

Step 4 of 5

Calculate the current $i_0(\infty)$.

$$\begin{aligned} i_0(\infty) &= \frac{35}{20} \\ &= 1.75 \text{ A} \end{aligned}$$

Step 5 of 5

Calculate the time constant τ .

$$\begin{aligned} \tau &= \frac{L}{R} \\ &= \frac{1}{2} \end{aligned}$$

The current i_0 across the inductor is,

$$\begin{aligned} i_0 &= i_0(\infty) + (i_0(0^-) - i_0(\infty))e^{\frac{-t}{\tau}} \\ &= 1.75 - 0.75e^{-2t} \text{ A} \end{aligned}$$

Hence, the current across the inductor is $\boxed{1.75 - 0.75e^{-2t} \text{ A}}$.